综合练习

人员

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上周作业检查

上上周作业链接: https://vjudge.net/contest/799964

Rank	Team	Score	Penalty	A 9 / 12	B 9 / 9	C 9 / 14	D 6 / 7	E 10 / 14	F 5 / 8	G 8 / 18	H 8 / 10	I 0 / 0
1	yyycj	8	30615	0:42:13	0:50:38	1:18:09	5:29:08	1:15:43:25	7:00:37:00	5:16:22:38	6:13:12:00	
2	chenxinmiao (陈欣妙)	8	31655	0:41:14	0:48:18	1:31:33 (-1)	1:14:01:00	2:13:43:00	7:00:47:53 (-1)	3:13:05:00 (-2)	7:01:37:02	
3	wcz123bc	8	68872	5:34:23	5:48:19	6:51:03	6:02:56:02	7:05:28:53	13:05:00:02	13:04:46:25 (-3)	7:06:07:13 (-1)	
4	r111	7	24387	5:31:24	5:54:08	6:21:03 (-1)	1:14:27:01	2:12:49:15		3:13:01:00	8:12:04:00	
5	wyb1111 (王毅博)	7	30873	5:41:00 (-1)	5:55:12	6:17:37	13:35:00 (-1)	7:05:35:00 (-1)		5:13:05:00 (-3)	7:06:05:00 (-1)	
6	xuruiqian	7	42560	5:42:09 (-1)	5:54:01	6:25:07	7:03:12:00	7:05:36:00		7:03:41:00	7:06:30:00	
7	junyan007008	7	66836	0:48:31 (-1)	1:18:35	10:12:51:59 (-1)		14:00:29:47 (-2)	7:07:16:00 (-1)	7:07:38:00 (-1)	7:01:34:00	
8	chujinxuan (褚锦轩)	4	10345	0:46:41	0:52:01	1:30:43 (-1)		7:00:55:42	+1			
9	lyc12345bc (刘奕辰)	4	67485					13:13:49:05	7:01:51:31 (-1)	13:01:52:59 (-1)	13:02:31:30	
10	lijinshu	3	204	0:45:32	0:53:14	1:25:29 (-1)						
11	7654Fox (隋天乙)	1	10193					7:01:33:10				

上周作业链接: https://vjudge.net/contest/801929

本周作业

https://vjudge.net/contest/803886 (课上讲了上周的 A ~ E 题, 课后作业是本周的 A ~ E 题)

课堂表现

大部分同学上周补题补得还是不错的, 希望同学们接下来继续保持。

课堂内容

CF1272D Remove One Element

f1[i] 代表以 i 为结尾时最长连续上升是多少, f2[i] 代表以 i 为开头时最长连续上升是多少

```cpp
#include <bits/stdc++.h>

using namespace std;

const int maxn = 2e5 + 5;
int w[maxn];
int f1[maxn], f2[maxn];

int main()
{
  int n; cin >> n;
  for (int i = 1; i <= n; ++i) cin >> w[i];

  int res = 0;
  for (int i = 1; i <= n; ++i) {
    if (w[i] > w[i-1]) f1[i] = f1[i-1]+1;
    else f1[i] = 1;
    res = max(res, f1[i]);
  }

  for (int i = n; i >= 1; --i) {
    if (w[i] < w[i+1]) f2[i] = f2[i+1]+1;
    else f2[i] = 1;
  }

  for (int i = 2; i <= n-1; ++i) {
    if (w[i-1] < w[i+1]) res = max(res, f1[i-1] + f2[i+1]);
  }
  cout << res << endl;
  return 0;
}
```

CF1551E Fixed Points

f[i][j]: 考虑前 i 个数, 如果删了 j 个时, 最多能有多少个数在对应的合适位置

```cpp
#include <bits/stdc++.h>

using namespace std;

const int maxn = 2000 + 5;
int w[maxn], f[maxn][maxn];

void solve() {
  int n, k; cin >> n >> k;
  for (int i = 0; i <= n+2; ++i) {
    for (int j = 0; j <= n+2; ++j) f[i][j] = 0;
  }

  for (int i = 1; i <= n; ++i) cin >> w[i];
```

```
  for (int i = 1; i <= n; ++i) {
    f[i][0] = f[i-1][0] + (w[i]==i);
    for (int j = 1; j < i; ++j) {
      f[i][j] = max(f[i-1][j-1], f[i-1][j] + (w[i]==i-j));
    }
  }

//  cout << "---------------------------- ";
  for (int i = 0; i < n; ++i) {
    if (f[n][i] >= k) { cout << i << endl; return; }
  }
  cout << -1 << endl;
}

int main()
{
  int T; cin >> T;
  while (T -- ) solve();
  return 0;
}
```

CF1288E Messenger Simulator

建一个 2*N 的数组, 用树状数组进行过程的模拟

```
#include <bits/stdc++.h>

using namespace std;

const int N = 3e5 + 5;
int up[N], down[N], f[N];

int tr[N*2];
int lowbit(int x) { return x&(-x); }
void update(int x, int k) {
  while (x < N*2) tr[x] += k, x += lowbit(x);
}
int query(int x) {
  int res = 0;
  while (x) res += tr[x], x -= lowbit(x);
  return res;
}

int main()
{
  int n, m; cin >> n >> m;
  for (int i = 1; i <= n; ++i) {
    up[i] = down[i] = i;
    f[i] = N+i, update(f[i], 1);
  }
```

```
  for (int i = 1; i <= m; ++i) {
    int x; cin >> x;
    down[x] = max(down[x], query(f[x])), up[x] = 1;
    update(f[x], -1);
    f[x] = N-i, update(f[x], 1);
  }

  for (int i = 1; i <= n; ++i) down[i] = max(down[i], query(f[i]));

  for (int i = 1; i <= n; ++i) cout << up[i] << " " << down[i] << endl;
  return 0;
}
```

CF1286B Numbers on Tree

先 dfs 求每棵子树的大小, 判定无解情况

如果有解, 可以自上往下处理进行填数

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 2000 + 5;
vector<int> vec[maxn];
int w[maxn], sz[maxn];

bool dfs(int u, int fa) {
  sz[u] = 1;
  for (int i : vec[u]) {
    if (i == fa) continue;
    if (!dfs(i, u)) return false;
    sz[u] += sz[i];
  }

  if (w[u] > sz[u]-1) return false;
  return true;
}

int f[maxn];
set<int> s;

void dfs_2(int u, int fa) {
  int cnt = 0, t;
  for (int i : s) {
    if (cnt == w[u]) { t = i; break; }
    ++cnt;
  }
  f[u] = t, s.erase(t);

  for (int i : vec[u]) {
```

```
    if (i == fa) continue;
    dfs_2(i, u);
  }
}

int main()
{
  int n; cin >> n;
  int root;
  for (int i = 1; i <= n; ++i) {
    int a; cin >> a >> w[i];
    if (!a) root = i;
    else vec[a].push_back(i);
  }

  if (!dfs(root, -1)) { cout << "NO" << endl; return 0; }

  cout << "YES" << endl;

  for (int i = 1; i <= n; ++i) s.insert(i);
  dfs_2(root, -1);
  for (int i = 1; i <= n; ++i) cout << f[i] << " "; cout << endl;
  return 0;
}
```

CF1856C To Become Max

二分, 看是否能 <=k 次把 w[pos] 变成 value

这个问题可以向后递归完成

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 1000 + 5;
int w[maxn];

bool dfs(int n, int pos, int k, int value) {  // 是否能 <=k 次把 w[pos] 变成 value
  if (k < 0) return false;
  if (w[pos] >= value) return true;
  if (pos == n) return false;

  if (dfs(n, pos+1, k-(value-w[pos]), value-1)) return true;
  return false;
}

bool check(int n, int k, int mid) {
  for (int i = 1; i <= n; ++i) {
    if (dfs(n, i, k, mid)) return true;
  }
```

```
    return false;
  }

  void solve() {
    int n, k; cin >> n >> k;
    for (int i = 1; i <= n; ++i) cin >> w[i];

    int l = 1, r = 2e8;
    while (l <= r) {
      int mid = (l + r) / 2;
      if (check(n, k, mid)) l = mid+1;
      else r = mid-1;
    }

  //  cout << "----------------- ";
    cout << r << endl;
  }

  int main()
  {
    int T; cin >> T;
    while (T -- ) solve();
    return 0;
  }
```